

FAQ

WAFT | Frequently Asked Questions

General Questions

What does WAFT stand for?

WAFT doesn't have an official acronym expansion. It's simply the name of the evolutionary code laboratory. Think of it as code "wafting" through evolutionary generations.

What is WAFT for?

WAFT is a framework for **directed evolution of AI agents**. Instead of manually engineering agents, you breed them through evolutionary pressure. The fittest agents survive and reproduce.

Who is WAFT for?

- AI researchers studying artificial cognition
- Developers who want self-improving agents
- Scientists needing rigorous evolution data
- Anyone curious about evolutionary AI

Technical Questions

What LLMs does WAFT support?

WAFT supports any LLM with a compatible API:

- OpenAI (GPT-4, GPT-3.5)
- Anthropic (Claude)
- Local models (via OpenAI-compatible APIs)
- Azure OpenAI
- Any OpenRouter model

How long does evolution take?

Depends on:

- Population size (10-100 agents typical)
- Generations (5-1000+)
- Gym complexity (simple scenarios: seconds; complex: minutes)

A typical 10-generation run with 20 agents takes 5-15 minutes.

Can I run WAFT locally?

Yes! Use a local LLM server (like ollama, llama.cpp, or vLLM) with OpenAI-compatible API. Set the base URL in configuration.

Scint Questions

What is a Scint?

A **Scint** (Semantic Contradiction/Inconsistency/Noise Token) is an ontological error — a point where the AI's output doesn't match reality or logical constraints.

Think of it as a “reality fracture” in the AI's response.

Why use Scints for fitness?

Scints provide a meaningful fitness signal:

- They measure actual failure modes
- They're automatically detectable
- They span multiple dimensions (syntax, logic, safety, factuality)
- They create genuine evolutionary pressure

Can I define custom Scint types?

Yes! Extend the `ScintDetector` class:

```
class MyScintDetector(ScintDetector):  
    def detect(self, text: str) -> list[Scint]:  
        # Your custom detection logic  
        ...
```

Evolution Questions

What is a “genome” in WAFt?

An agent's genome includes:

- Source code (the agent class)
- Configuration (parameters, settings)
- System prompts

Changes to any of these are mutations.

How does breeding work?

Two parent agents combine:

1. Code sections may swap
2. Configurations merge
3. Prompts may blend

The offspring inherits traits from both parents, with possible mutations.

What prevents degenerate evolution?

Several mechanisms:

- Minimum fitness thresholds
- Diversity maintenance
- Elitism (best agents always survive)
- Multi-objective fitness

Teleport Massive Questions

Is Teleport Massive real?

Teleport Massive is a **fictional corporation** within the WAFT universe. It provides narrative context for the framework and makes documentation more engaging.

What is the connection between WAFT and Teleport Massive?

In the fiction, WAFT is a scientific instrument developed by Teleport Massive for AI research. The Scint system emerged from their quantum teleportation research.

What is the Quantum Incident?

A fictional event where researcher Sarah Chen experienced multi-state existence during a teleportation test, leading to the discovery of Scints.

Practical Questions

How do I get started?

```
uv tool install waft
waft new my_lab
cd my_lab
waft verify
waft evolve --agent BaseAgent --generations 5
```

Where can I get help?

- GitHub Issues: <https://github.com/ctavolazzi/waft/issues>
- Documentation: <https://github.com/ctavolazzi/waft>
- Discord: [Coming soon]

How do I contribute?

1. Fork the repository
2. Create a feature branch
3. Submit a pull request
4. Follow the contribution guidelines

Still have questions?

Open an issue on GitHub or check the documentation.

“Don’t just build agents. Breed them.”